micro:bit

Introduction to Computer Science

Unit 12: Final project

Microsoft

Lesson A
Looking back

What we'll learn in Lesson A

- A review of the computer science concepts we have covered
- A chance to brainstorm

Project overview

You will use the design thinking approach to create an original project that serves a purpose by:

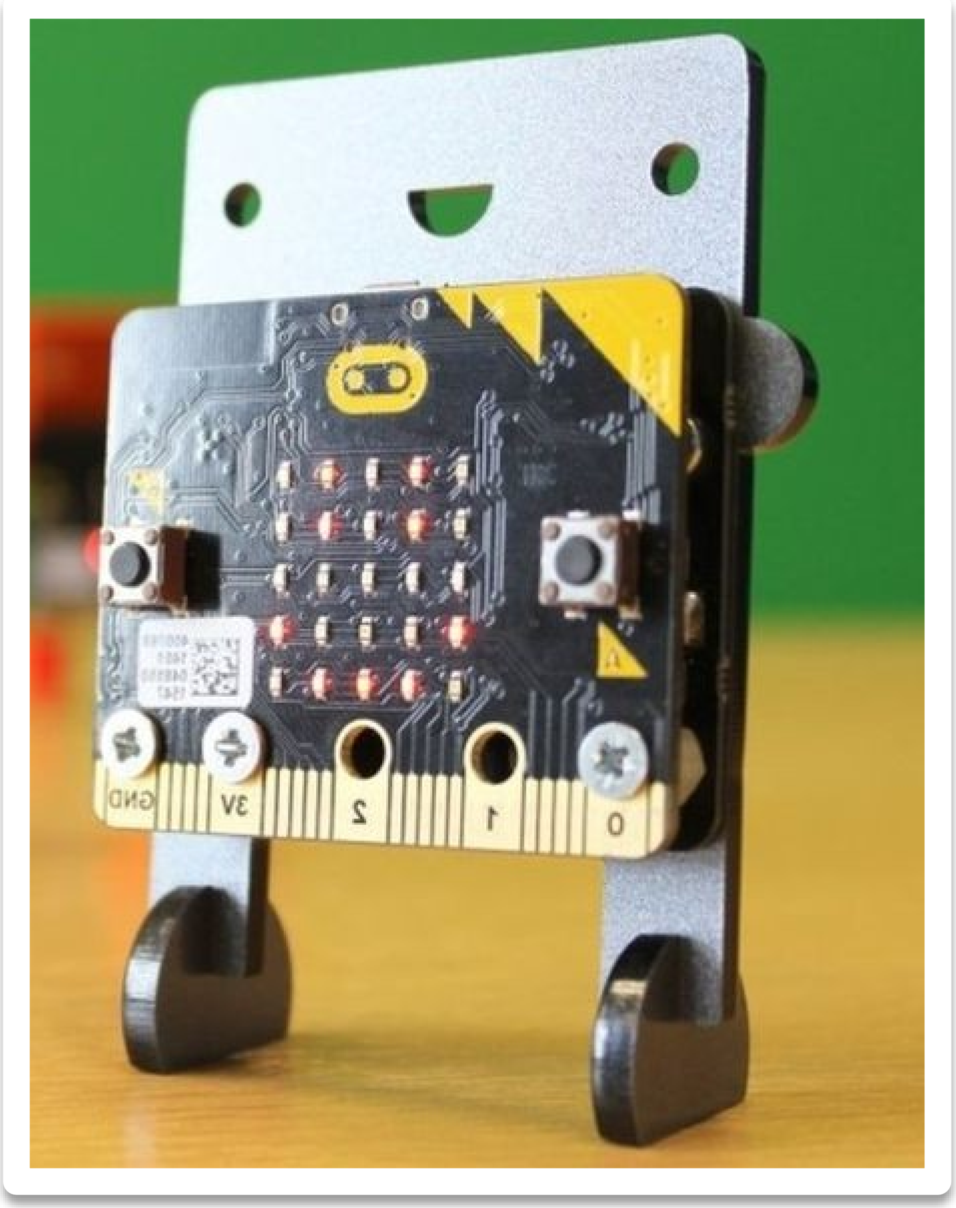
- Solving a problem
- Filling a need

Show what you know

- Learn and demonstrate something new
- Include a maker component

Demonstrate the use of the following concepts:

  - Coordinates
  - Booleans
  - Bits, bytes, and binary
  - Radio communications
  - Arrays



Concepts review

1. Coordinates
2. Booleans
3. Bits, bytes and binary
4. Radio communications
5. Arrays
6. Accelerometer

Name one thing you've learned from each of these lessons.

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Coordinates

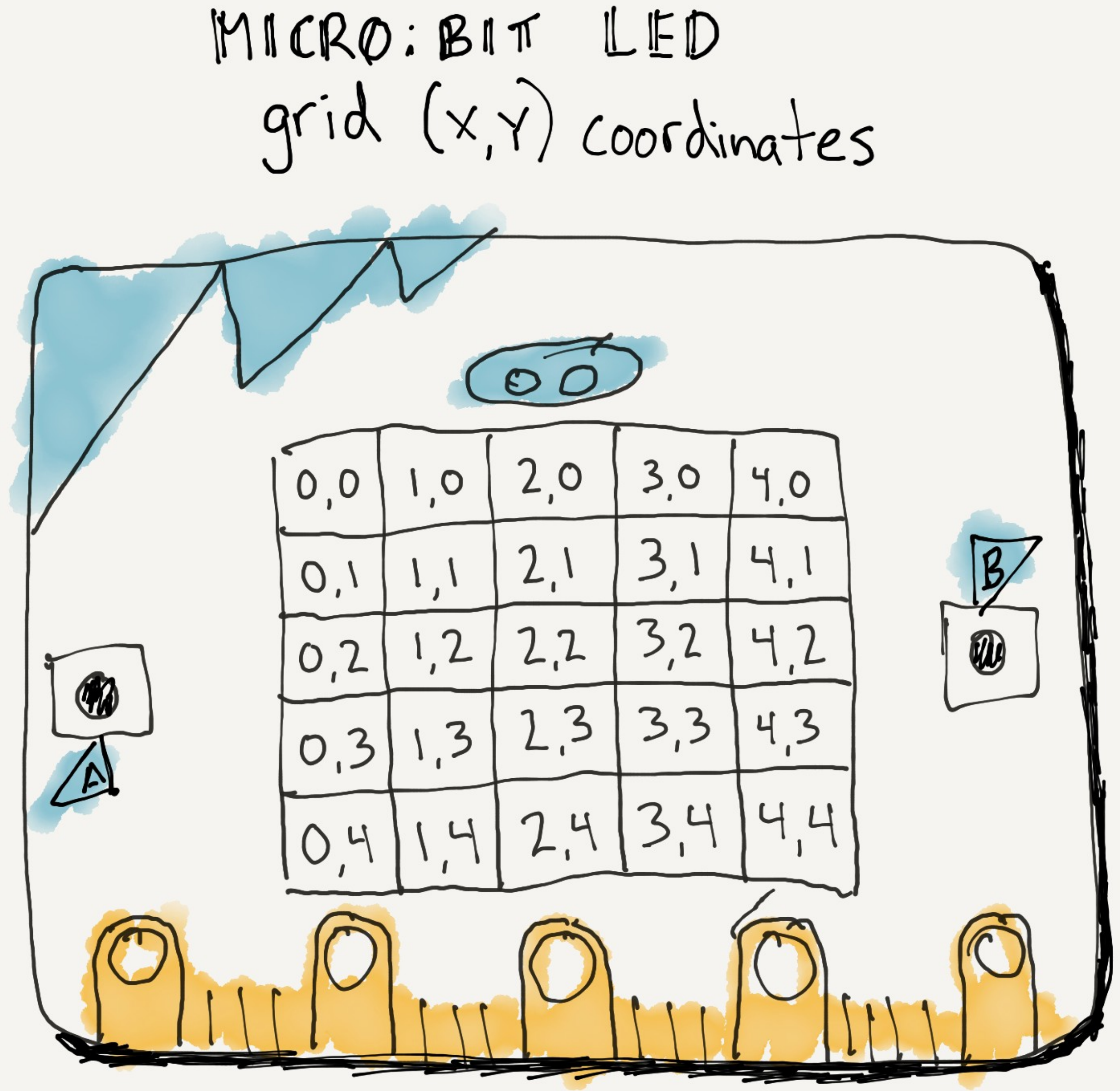
The 5 x 5 grid of LEDs on the micro:bit represents a coordinate grid with a **horizontal x-axis** and a **vertical y-axis**.

The **origin (0,0)** is in the **top-left corner**.

The values of the y coordinates range from **0 through 4** and increase from **top to bottom**.

Individual LEDs can be turned on and off through the **LED Plot** and **Unplot** blocks.

The current **value** of an LED can be checked, and its **brightness** can be changed, as well.



Booleans

A **Boolean** is a data type that only has two possible values: **True or False**.

You can use boolean variables to **keep track of the state of a game** (**gameOver** is either true or false) or check to see **whether a certain action has taken place** yet (**messageSent** is either true or false).

Boolean operators such as **AND, OR,** and **NOT** allow you to **combine boolean expressions** to make more complex conditions.

Bits, bytes and binary

Computers work with the **base-2 number system** – **each place value is twice the previous one**.

Binary numbers only have **two possible values**: **0 or 1**.

A **bit** is the **smallest unit of information**, a bit can be **either 0 or 1**.

A **byte** is a **sequence of eight bits**.

A **bit** can be used to **hold a Boolean value** where:

- **0 = false**
- **1 = true**

Radio communications

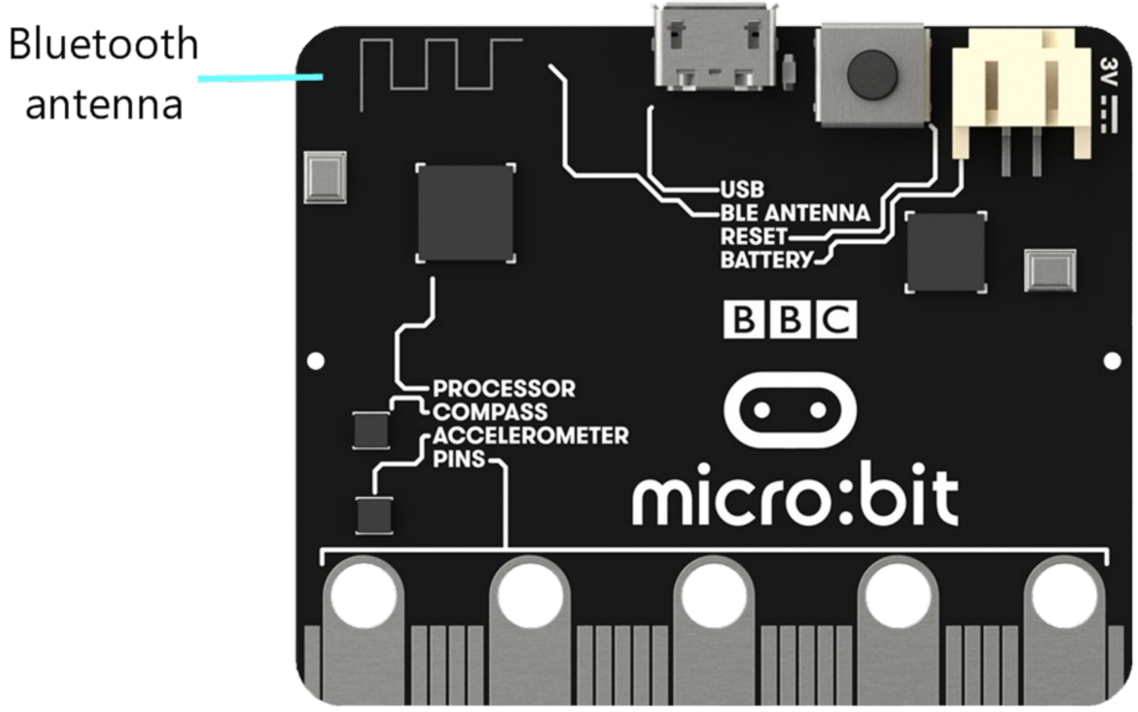
Radios use **electromagnetic waves** to send messages through the air.

A radio communication system requires a **transmitter** to send messages, and a **receiver** to receive messages.

On the back of the micro:bit, if you look closely in the upper left corner, there is an **antenna**, which is how the micro:bit can send and receive messages through radio waves.

You can send a **number, string** or **string/number combination** in a message.

You can give a micro:bit instructions on what to do when it receives a radio message.



Arrays

An **array** is a **list, or collection, of similar things**.

Arrays in MakeCode are used to **store and retrieve numbers, strings, musical notes,** or **sprites**.

Every spot in an array can be identified by its index. **The first spot in an array is always index 0**.

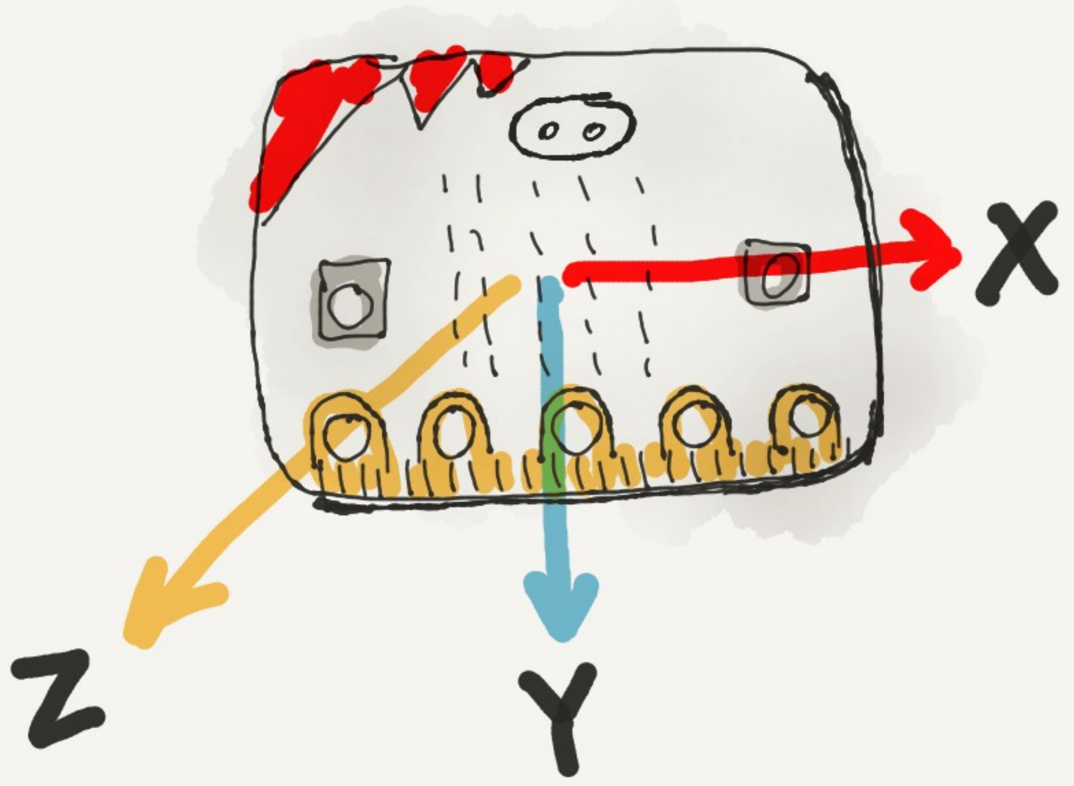
Objects can be **accessed, changed, added to, or removed from an array** using their index.

Three common methods of sorting elements in an array are **bubble sort, selection sort,** and **insertion sort**

Accelerometer

The **accelerometer** detects movement along **three separate axes of movement**

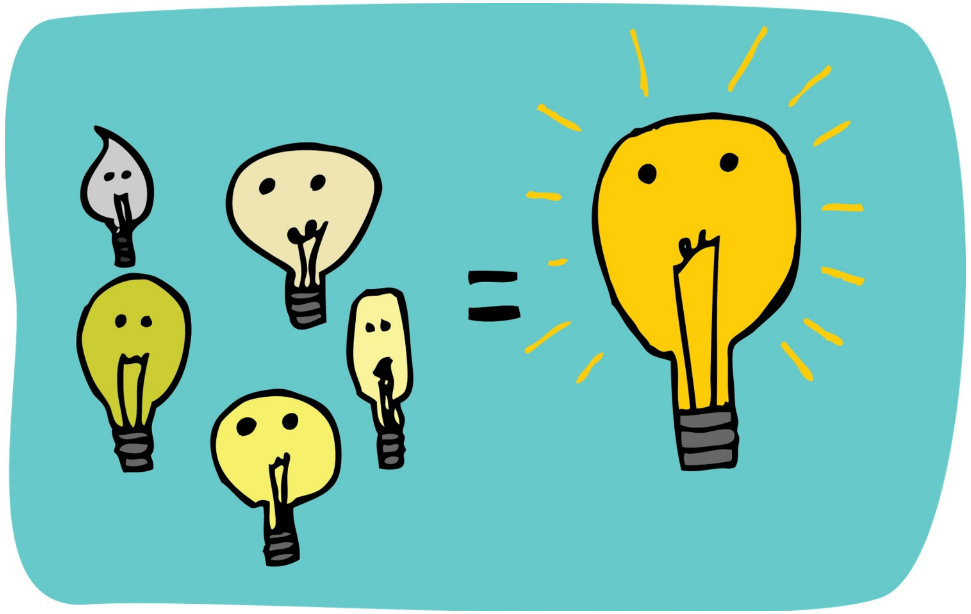
- **x-axis:** Tilt left/right
- **y-axis:** Tilt forward/back
- **z-axis:** Move up/down



face up on table

Final project brainstorm

- You will be creating a final project with the micro:bit that **incorporates the concepts** we've covered so far in class.
- Get in **groups of 4** to **brainstorm** ideas for your final projects. Think about all the concepts we've covered so far and the projects you've created.
- Your final project can build off an existing project or be something brand new.



What we learned in Lesson A

- A fresh look at the computer science concepts we have covered
- A chance to brainstorm for your final project!

Next time: Lesson A

- Code and make a final project

Lesson B

Coding and making a final project

What we'll learn in Lesson B

- Design a unique project and build it!

Final project

Design a project by yourself that serves a purpose by:

- Solving a problem
- Filling a need

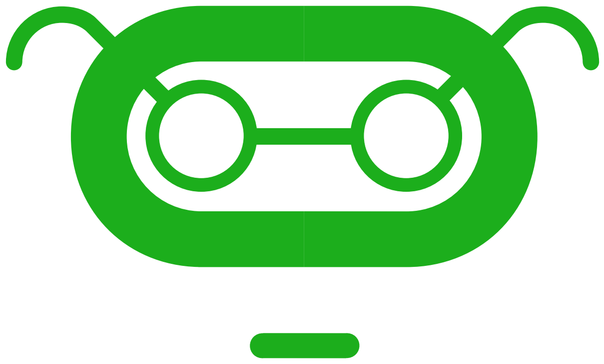
Show what you know:

- Learn and demonstrate something new
- Include a maker component

Project expectations

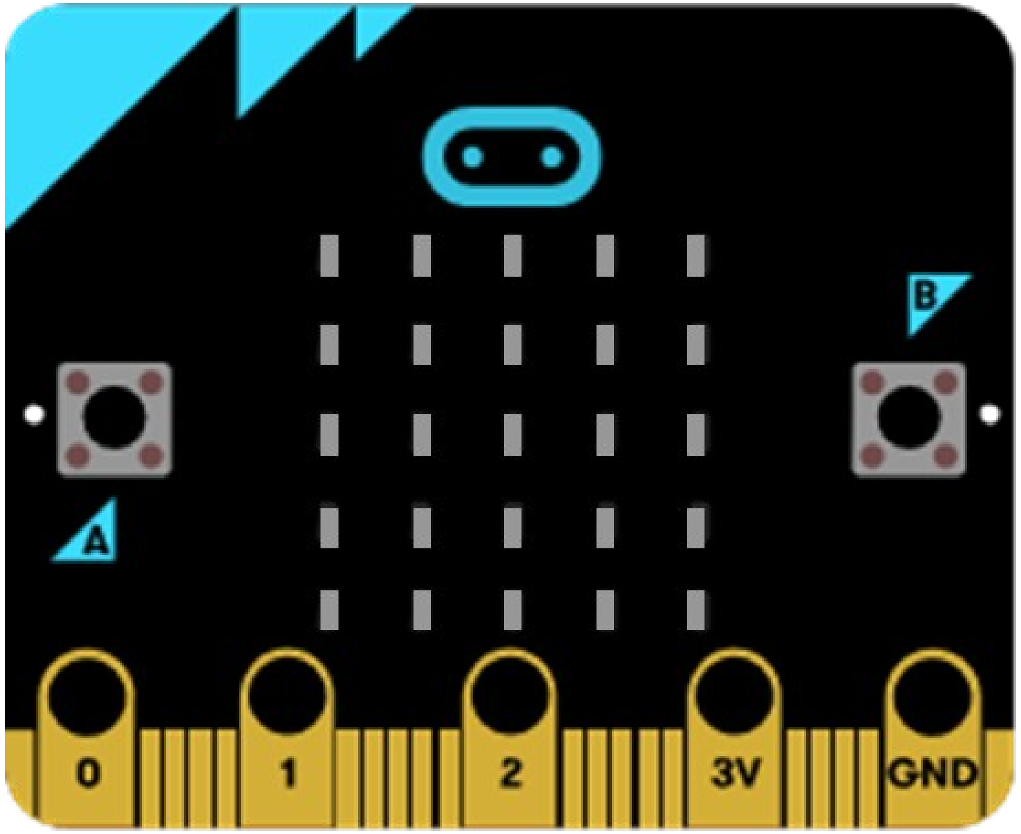
Use a **design thinking approach** and ensure the project meets these specifications:

- Create an **original project using the micro:bit.**
- Incorporate a **physical component** to the project.
- Demonstrate the use of the following **concepts**:
  - **Coordinates**
  - **Booleans**
  - **Bits, bytes, and binary**
  - **Radio communications**
  - **Arrays**
  - **Accelerometer**



Project ideas

- Create a game
- Create something that helps somebody by solving a problem
- Create something beautiful
- Create a musical instrument



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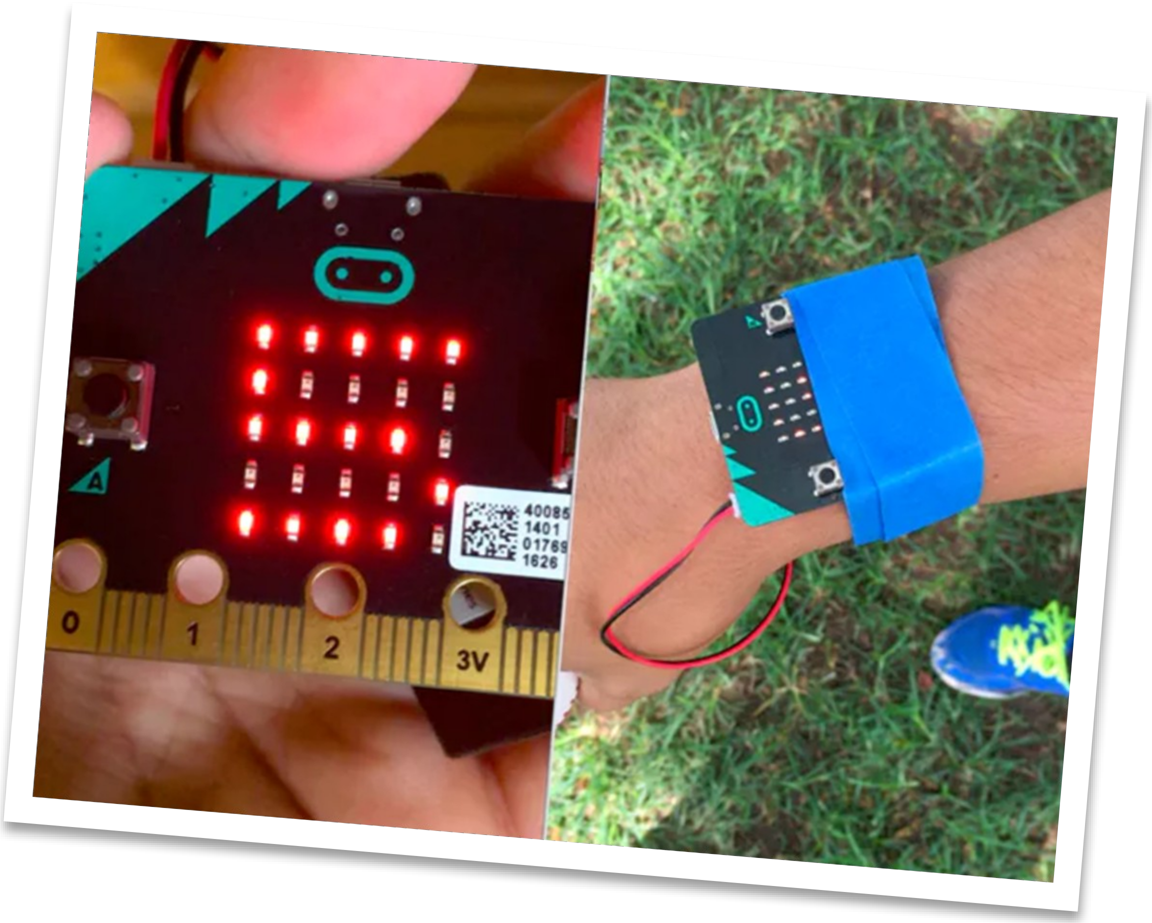
Project example: Baseball pitch counter

This project straps to a pitcher's arm and uses the micro:bit accelerometer to record how many pitches have been thrown in a session.



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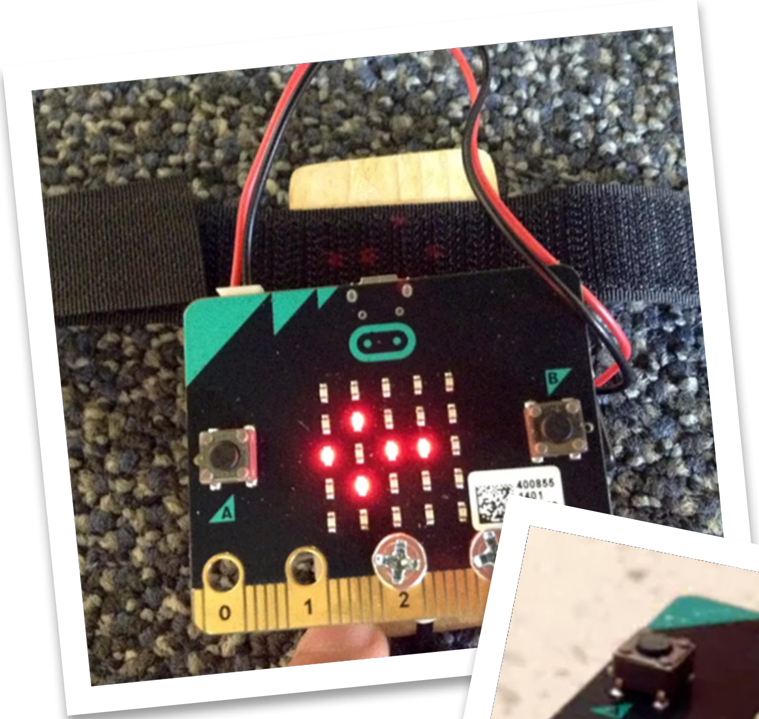
Project example: Wrist-mounted step counter and compass

This project straps to your wrist, displays a compass that updates as you walk around and keeps track of your steps.

The micro:bit is elevated to allow room for the battery pack to fit underneath.



Project example: Violin tuner

This project uses a piece of cardboard to mount the micro:bit to the side of a violin.

This student wanted to use it to tune his violin by playing a specific series of tones.

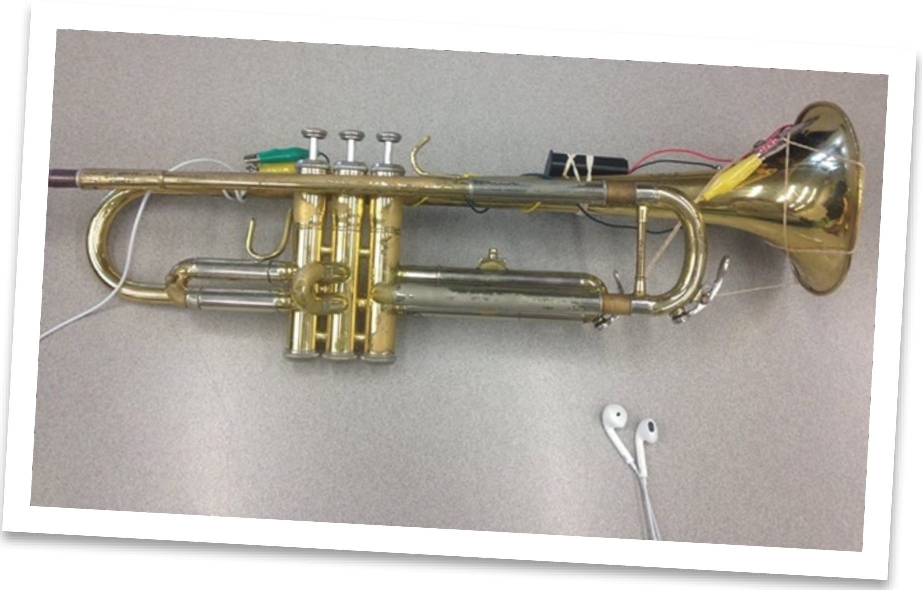
The micro:bit displays the note being played.

Project example: Trumpet angle detector

This example was used for marching band practice, where students must hold the trumpet at a 15-degree angle to avoid hitting the person in front of them or playing directly into their ears.

Because the trumpet is heavy, new trumpet players tend to let the trumpet droop.

This displays an icon (a check mark or an X) to help new trumpet players learn what the proper angle is supposed to feel like.



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What we learned in Lesson B

- Designed a project and built it!

Next time: Lesson C

- Final project showcase

Lesson C

Final project showcase

What we'll learn in Lesson C

- The amazing inventions everyone made!

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Showcase

- Give a demo of how it works
- Show how the code works
- Show how you assembled the physical/maker component
- Tell us how you used one concept you knew
- Tell us how you learned something new

Final narrative

- How did you start the process of designing the product/meeting your goals?
- What did you hope to learn?
- What challenges did you face? How did you overcome them?
- What was the outcome?
- What did you learn in the end?
- Who in the class provided help to you along the way? How?
- What were you proud of?
- What would you do differently next time?

Sample final narrative

It's clear to me now that in the second week, I was a little lost and confused with the direction my project was taking. I can see now that in my chats with Ms. Kiang and with classmates (Conference Notes 4/3), I was not being very precise in my questions, and I didn't totally understand what she was explaining.

Looking back, one of my goals was to meet more regularly with my table mates. Hopefully, I would be a lot less confused and at least this time, when I got stuck, we would be able to solve it together. I wrote about this in my Record of Thinking (Record of Thinking #2) but I am surprised that things cleared up for me so quickly once we did start meeting together. This allowed me to get past something that was really bothering me—specifically adding and removing things from an array—and I was able to complete that in less than a day after having been stuck for more than a week (Work Log #4).

Once I started to get a little more clear on what to do, I was able to get more effective help from my classmates. Specifically, Jordan helped me a lot with figuring out how to get an image to display properly on the screen. He also showed me how to search through the online documentation more effectively. I think if I could do this over again, I would have scheduled more time earlier to meet with Ms. Kiang and/or found a better way to share the different online sites with my table mates, because we all found different places to go. I didn't even find out until the end that you could jump into JavaScript to make changes to the code, and it makes it all with the right blocks when you go back (Beta Testing notes)! That would have saved me a lot of time.

What we learned in Lesson C

- That we learned a lot!
- That we're all very creative!

Thank you! :)